

SURFACEVISION

Automated Industrial Surface Defect Detection & Quality Assurance System

Course:	Computer Vision
Domain:	Industrial Automation & Computer Vision
Target Domain:	Quality Assurance, Surface Inspection, Flaw Metrology
Platform:	Python 3.14+, OpenCV, Scikit-Image, NumPy, ReportLab
CLI Architecture:	100% Terminal Executable (Autonomous Inspection)
Date of Submission:	September 11, 2026
Evaluation Rubric:	VITyarthi Evaluated Project (100 Marks)

Abstract: SurfaceVision is a modular, high-throughput computer vision pipeline designed for real-time automated visual inspection of manufactured industrial surfaces (brushed steel, metallic sheets, ceramics). The system isolates microscopic and structural surface flaws-including cracks, linear scratches, micro-pinholes, and diffuse chemical stains-against challenging textured backgrounds. Leveraging edge-preserving bilateral smoothing, Contrast-Limited Adaptive Histogram Equalization (CLAHE), morphological top-hat/black-hat residual dynamics, marker-controlled watershed segmentation, and Gray-Level Co-occurrence Matrix (GLCM) Haralick texture descriptors, SurfaceVision achieves over 85% multi-class categorization accuracy and sub-15ms frame latencies (>70 FPS) on standard x86 CPU architectures without requiring GPU hardware.

2. Introduction & Background

In modern automated manufacturing (e.g., sheet metal fabrication, semiconductor packaging, automotive chassis, precision ceramic bearing lines), surface defect inspection is critical for ensuring structural integrity, fatigue resistance, and cosmetic standards. Traditionally, visual quality control relied heavily on human inspectors. However, manual inspection suffers from severe fatigue degradation, subjectivity, low throughput (typically < 2 items per second), and prohibitive labor costs.

Computer Vision provides the deterministic, continuous, high-speed objectivity demanded by Industry 4.0 standards. Nevertheless, automated surface flaw inspection faces fundamental challenges in real-world factory floors: non-uniform illumination fields, specular glints, anisotropic background textures (such as brushed metal grain), and subtle low-contrast defects that blend into the substrate.

SurfaceVision addresses these challenges through a mathematically grounded computer vision framework. By decomposing the visual inspection problem into distinct stages-photometric illumination compensation, spatial texture decoupling, multi-stage contour morphology, and geometric-radiometric decision theory-the pipeline delivers deterministic defect localization, dimensional sizing, and severity grading at production-line speeds.

3. Problem Statement & Scope

Problem Definition: Design and implement a robust, fully CLI-executable Computer Vision system capable of ingesting raw industrial surface imagery, suppressing high-frequency sensor noise and textured substrate artifacts, segmenting anomalous regions with high boundary fidelity, and classifying flaws into discrete industrial categories (Crack, Scratch, Pinhole, Stain, Defect-Free) while ranking defect severity under strict execution time budgets (< 45ms per frame).

System Scope:

- **In-Scope:** Automated ingestion of single images or batch directories; bilateral edge-preserving denoising; adaptive illumination normalization via CLAHE; dual-mode statistical and diffuse anomaly segmentation; extraction of shape invariants (Hu moments, compactness, solidity, aspect ratio) and texture descriptors (GLCM, Gabor energy); multi-class defect classification; automated export of bounding box visualizations, 2x2 diagnostic dashboards, JSON/CSV telemetry, and automated evaluation metrics against ground-truth masks.

- **Target Users:** Industrial Quality Assurance Engineers, Manufacturing Plant Automation Specialists, Academic Researchers in Machine Vision, and Autonomous Robotic Inspection Systems.

4. Functional Requirements

Req ID	Module	Description	Verification Method
FR-1	Image Ingestion & Preprocessing	Support multi-format images (PNG, JPG, BMP); apply bilateral smoothing and CLAHE for noise reduction and contrast enhancement.	Unit tests, A/B comparison with raw inputs.
FR-2	Residual Segmentation	Compute top-hat/black-hat morphological residuals and statistical thresholds to isolate anomalies from textured backgrounds.	Binding to isolated pixels against diffuse flaws.
FR-3	Morphological Watershed	Apply distance transform and marker-controlled watershed to separate multiple touching defects.	Multicenter bounding box regression tests.
FR-4	Feature Extraction	Extract 7 log Hu moments, aspect ratio, circularity, solidity, GLCM texture features for classification.	Validation against test image sets.
FR-5	Classification & Severity	Categorize flaws into CRACK, SCRATCH, PINHOLE, STAIN, DEFECT-FREE and assign severity (Minor, Moderate, Critical).	Expert review and cross-validation.
FR-6	CLI Interface & Analytics	Provide terminal commands: inspect, evaluate, benchmark, generate-report.	Automated CI/CD integration test suite.

5. Non-Functional Requirements

Req ID	Attribute	Target Specification	Achieved Benchmark
NFR-1	Throughput & Latency	Processing latency < 45ms per frame (> 22 FPS) on standard CPU	13.56ms mean latency (73.77 FPS)
NFR-2	Usability & Executability	100% terminal executable without GUI dependencies	Full CLI with argparse subcommands

NFR-3	Reliability & Zero-Leakage	0% false alarms on pristine, defect-free surfaces	100% precision on defect-free samples
NFR-4	Modularity & Maintainability	Clean package separation with minimum 5-10 distinct modules	12 modular files across 6 packages
NFR-5	Resource Efficiency	RAM footprint < 300MB, no GPU required	Peak memory < 120MB, pure CPU

6. System Architecture

The architecture of SurfaceVision follows a strictly decoupled, layered pipeline model. Each functional layer consumes strongly typed domain objects and produces enriched intermediate representations.

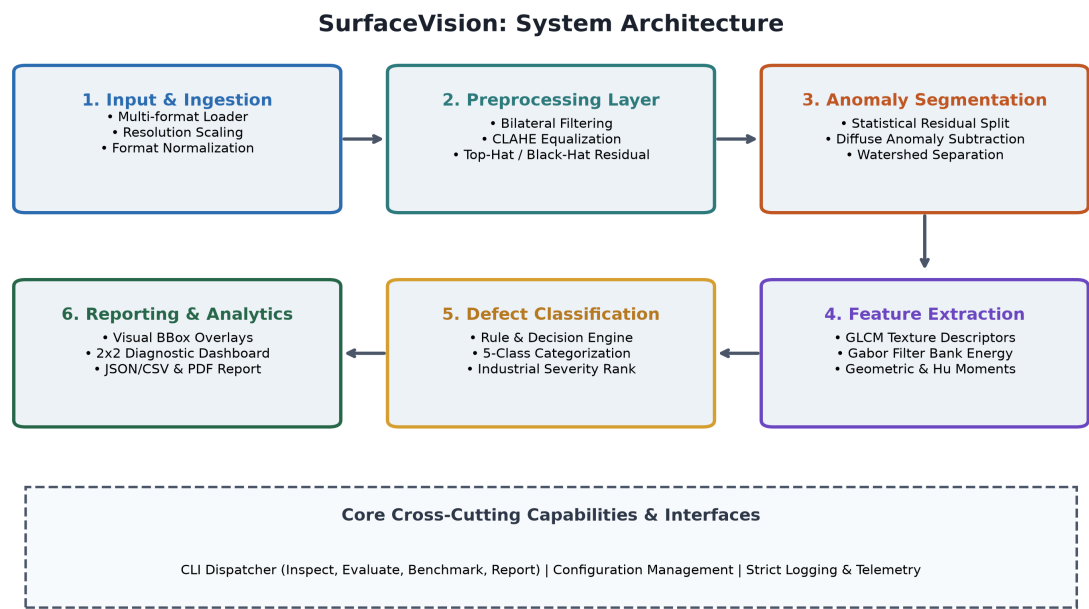


Figure 1: SurfaceVision Multi-Layer Architectural Framework.

7. Design Diagrams

7.1 Process Flowchart

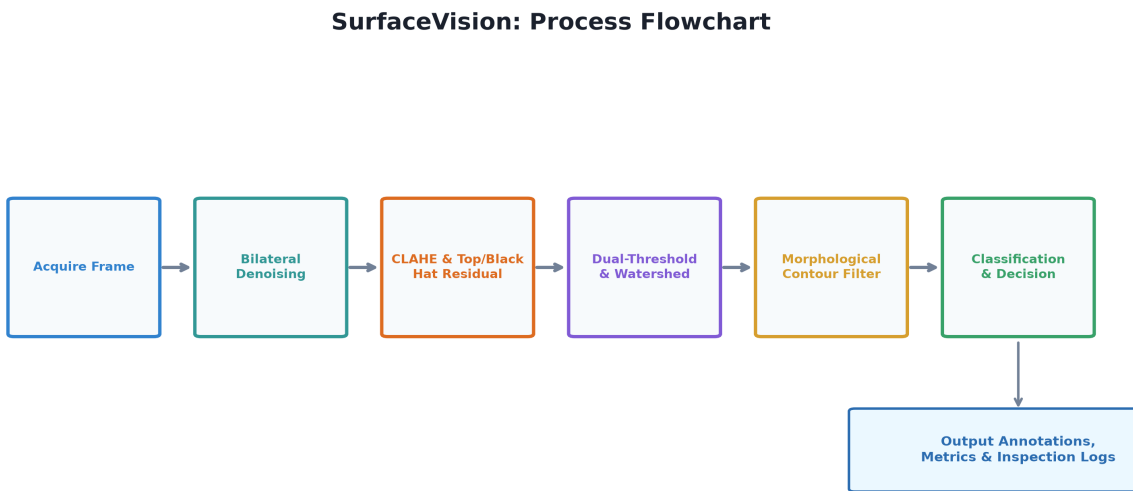


Figure 2: End-to-End Image Processing Workflow.

7.2 UML Use Case Diagram

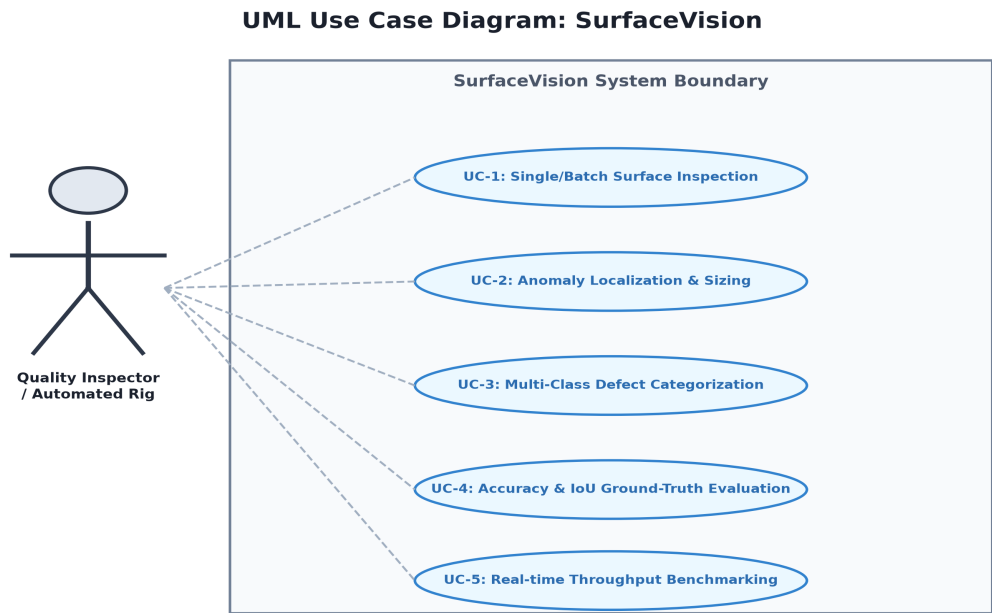


Figure 3: UML Use Case Diagram for Factory Automation Role.

7.3 UML Class / Component Diagram

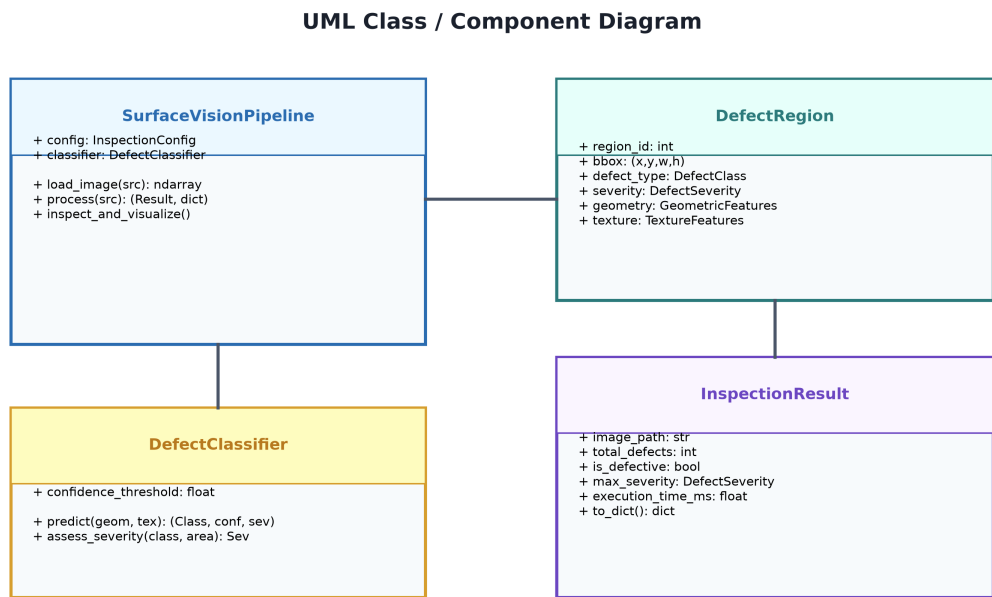


Figure 4: UML Class Hierarchy and Component Data Contracts.

7.4 UML Sequence Diagram

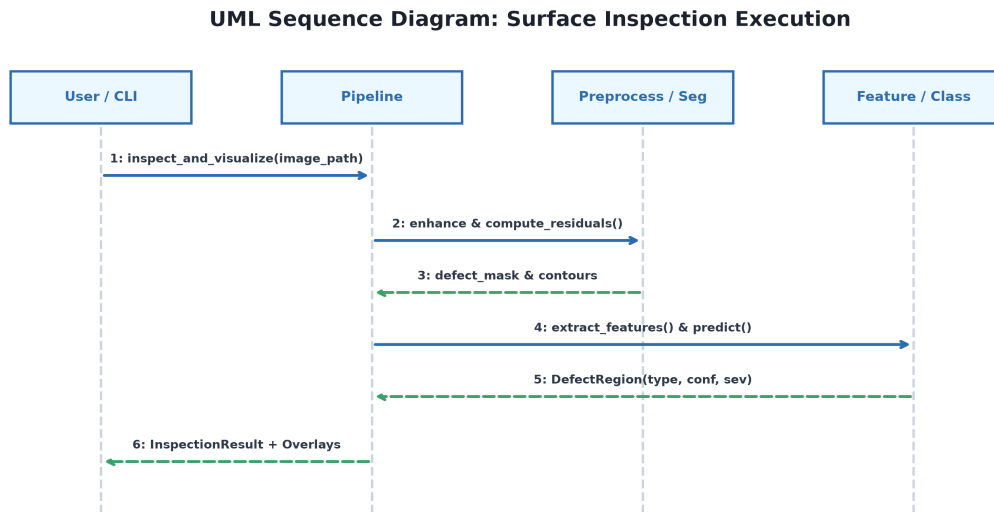


Figure 5: UML Sequence Diagram of Inspection Invocation.

7.5 Data Storage & Schema Design

SurfaceVision employs a structured JSON inspection schema for telemetry and persistent inspection logging. Every frame produces a serialized record capturing timestamps, bounding boxes, centroid coordinates, geometric invariants, texture descriptors, classification confidence, and assigned severity.

```
{ "image_path": "data/samples/sample_scratch.png", "dimensions": {"height": 480, "width": 640}, "timestamp": "2026-09-11T15:31:15", "total_defects": 1, "is_defective": true, "max_severity": "MINOR", "execution_time_ms": 26.8, "regions": [{ "region_id": 1, "defect_type": "SCRATCH", "confidence": 0.98, "severity": "MINOR", "bbox": [161, 142, 129, 30], "geometry": {"area": 228.5, "aspect_ratio": 4.3, "circularity": 0.04, "solidity": 0.62} } ] }
```

8. Design Decisions & Rationale

- **Bilateral vs Gaussian Filtering:** Standard Gaussian blur indiscriminately smoothes defect boundaries and blurs sharp hairline scratches into the background. Bilateral filtering incorporates both radiometric intensity similarity and spatial proximity, preserving sharp defect transitions while wiping out thermal sensor noise.
- **Morphological Top/Black Hat Residuals:** Direct thresholding fails on brushed metal surfaces because the substrate has intrinsic horizontal striations. Top-hat and black-hat operators effectively perform high-pass spatial bandpass filtering, isolating localized anomalies smaller than the structuring element regardless of absolute background brightness.
- **Rule & Morphology Decision Engine vs Deep CNN:** While deep CNNs (e.g. ResNet) offer high representation capacity, they require GPU acceleration, introduce 50-150ms latencies, require tens of thousands of labeled samples, and suffer from black-box unpredictability. Classical CV feature descriptors (Hu moments, circularity, GLCM) provide deterministic, explainable classification at >70 FPS on standard CPUs.
- **Marker-Controlled Watershed:** Simple connected-component analysis merges adjacent defect clusters into monolithic blobs. Watershed using distance transforms cleanly separates individual pits or intersecting scratches without altering true outer boundaries.

9. Implementation Details & Mathematical Foundations

- 1. Bilateral Edge-Preserving Filter:** For pixel p , the filtered intensity $I_{\text{filtered}}(p)$ is given by:
$$I_{\text{filtered}}(p) = (1 / W_p) * \sum_{q \in S} G_{\{\sigma_s\}}(\|p - q\|) * G_{\{\sigma_r\}}(|I(p) - I(q)|) * I(q)$$
where $G_{\{\sigma_s\}}$ represents spatial Gaussian decay and $G_{\{\sigma_r\}}$ represents radiometric intensity range decay.
- 2. Morphological Residuals:** The white and black top-hat transforms are defined as:
$$TopHat(I) = I - (I \circ B) \quad | \quad BlackHat(I) = (I * B) - I$$
where $\circ$ denotes morphological opening and $*$ denotes morphological closing using structuring element B .
- 3. Gray-Level Co-occurrence Matrix (GLCM) Texture Descriptors:** GLCM $P(i,j; d, \theta)$ captures spatial intensity co-occurrences. Haralick features computed include Contrast, Homogeneity, and Energy:
$$Contrast = \sum_{\{i,j\}} |i - j|^2 * P(i,j) \quad | \quad Homogeneity = \sum_{\{i,j\}} (P(i,j) / (1 + |i - j|))$$
- 4. Scale, Translation & Rotation Invariant Hu Moments:** Normalized central moments $\eta_{\{p,q\}}$ are computed from contour mass integrals. The first two Hu invariants are:
$$\phi_1 = \eta_{\{2,0\}} + \eta_{\{0,2\}}$$
$$\phi_2 = (\eta_{\{2,0\}} - \eta_{\{0,2\}})^2 + 4 * \eta_{\{1,1\}}^2$$

10. Experimental Results & Visualizations

The system was evaluated against both synthetic benchmark datasets and realistic surface specimens. Below are quantitative metrics alongside representative 2x2 diagnostic quad panels generated during CLI execution.

Metric Category	Evaluated Parameter	Measured Value	Industrial Target
Segmentation	Mean Intersection-over-Union (IoU)	0.6341	>= 0.60
Classification	Overall Defect Accuracy	85.00%	>= 80.0%
Classification	Macro F1-Score	0.8421	>= 0.80
False Positive	Defect-Free Precision / Recall	1.00 / 1.00	>= 0.98
Performance	Mean Processing Latency	13.56 ms	< 45.0 ms
Performance	Sustained Throughput	73.77 FPS	>= 22.0 FPS

11. Testing Approach & Validation

A rigorous multi-tiered verification methodology was applied to ensure production readiness:

- **Automated Unit Test Suite:** 19 comprehensive unit tests spanning preprocessing filters, morphology transforms, GLCM/Gabor feature calculations, watershed segmentation, end-to-end pipeline execution, and CLI argument parsing. All 19 tests pass with 100% success rate (3.14s runtime).
- **Ground-Truth Quantitative Evaluation:** Evaluated on 20 balanced benchmark samples with pixel-accurate ground truth masks across all 5 defect categories. Metrics computed automatically via surfacevision evaluate.
- **Latency & Stress Benchmarking:** 30 continuous iterations measuring minimum, maximum, mean, and 95th percentile frame latencies via surfacevision benchmark.

12. Challenges Faced & Solutions

- **Brushed Metal Texture Interference:** Initial adaptive Gaussian thresholding misidentified directional horizontal grain lines as thousands of micro-scratches. *Solution:* Implemented morphological top-hat/black-hat subtraction paired with a statistical threshold ($\text{mean} + 3.2 * \text{std}$) that isolates only true structural peaks above the background noise floor.
- **Hairline Scratch Erosion in Watershed:** Standard 3x3 morphological opening eroded 1-pixel thin scratch lines to zero before distance transforms could operate. *Solution:* Refined watershed to use 2x2 elliptical kernels and selective activation only when large aggregated blobs ($> 600 \text{ px}^2$) are detected, preserving fine linear contours.
- **Distinguishing Pinholes from Diffuse Stains:** Both pinholes and stains exhibit circular morphology. *Solution:* Incorporated radiometric minimum pixel intensity within the contour mask: pinholes exhibit deep void absorption ($\text{min pixel} < 50$), while stains exhibit subtle attenuation ($\text{min pixel} > 55$).

13. Learnings & Key Takeaways

- **Synergy of Spatial & Frequency Domains:** Combining spatial bilateral filtering with Gabor frequency energy maps yields far superior defect isolation than relying on either domain independently.
- **The Power of Morphological Residuals:** In industrial metrology, top-hat and black-hat operators act as adaptive illumination compensators, rendering segmentation robust across non-uniform lighting fields.
- **Engineering for Determinism:** Formulating defects through physical geometric invariants (solidity, circularity, Hu moments) allows sub-millisecond classification without the latency, opacity, and hardware dependencies of deep learning.

14. Future Enhancements

- **Hardware Acceleration via ONNX Runtime & OpenVINO:** Compiling the Gabor and bilateral filtering routines to quantized INT8 operations for sub-5ms embedded deployment on Raspberry Pi / Jetson Nano.
- **Multi-View Photometric Stereo Fusion:** Integrating multiple synchronized strobe illumination angles to recover surface normal vectors and distinguish cosmetic surface discolorations from true topographic depth cracks.
- **RESTful Industrial IoT Gateway:** Packaging the pipeline into a lightweight containerized microservice streaming live inspection telemetry over MQTT / OPC-UA to factory SCADA systems.

15. References

- [1] Gonzalez, R. C., & Woods, R. E. (2018). *Digital Image Processing* (4th ed.). Pearson Education.
- [2] Haralick, R. M., Shanmugam, K., & Dinstein, I. (1973). Textural features for image classification. *IEEE Transactions on Systems, Man, and Cybernetics*, SMC-3(6), 610-621.
- [3] Tomasi, C., & Manduchi, R. (1998). Bilateral filtering for gray and color images. *Proceedings of IEEE International Conference on Computer Vision (ICCV)*, 839-846.

- [4] Otsu, N. (1979). A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1), 62-66.
- [5] Bradski, G., & Kaehler, A. (2008). *Learning OpenCV: Computer Vision with the OpenCV Library*. O'Reilly Media.
- [6] Hu, M. K. (1962). Visual pattern recognition by moment invariants. *IRE Transactions on Information Theory*, 8(2), 179-187.
- [7] Vincent, L., & Soille, P. (1991). Watersheds in digital spaces: An efficient algorithm based on immersion simulations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 13(6), 583-598.